

After Alter Table

DESC customer;

SSN	Varchar (20)
Name	Varchar (100)
Email	Varchar (100)
Phone	Varchar (15)
Address	Varchar (200)

Task 2.1 Generating design of other traditional database model

AIM:

To perform DDL Data Definition Language and DML Data Manipulation Language commands

DDL commands

Create

Alter

Drop

Truncate

Rename

DML commands

Insert

Update

Delete

Select

DDL commands

1. CREATE TABLE

used to create a new table in the database

```
CREATE TABLE customer (
```

```
    SSN VARCHAR(20),
```

```
    Name VARCHAR(100),
```

```
    Email VARCHAR(100),
```

```
    Phone VARCHAR(15),
```

```
);
```

2. ALTER TABLE

used to add, delete or modify columns in an existing table

```
ALTER TABLE customer ADD Address VARCHAR(200);
```

After insertion

SSN	Name	Email	Phone	Address
CUST001	John	john@gmail.com	9876543210	New York
CUST002	Alice	alice@gmail.com	9123456780	miami

After update

SELECT * FROM CUSTOMER

SSN	Name	Email	Phone	Address
CUST001	John	john@gmail.com	900001234	New York
CUST002	Alice	alice@gmail.com	9123456780	miami

After Delete

SELECT * FROM CUSTOMER

SSN	Name	Email	Phone	Address
CUST001	John	john@gmail.com	900001234	New York

DESCRIBE OR DESC

Displays the structure of a table (column names and data types)

3. DROP TABLE

Deletes the entire table structure and all its data

DROP TABLE customer;

4. Truncate

Truncate command is used to remove all data but keep structure

Truncate TABLE customer;

DML Commands

DML commands are used to manage and manipulate data inside database tables

1. Insert

Insert new rows into a table

```
INSERT INTO customer (SSN, Name, Email, Phone, Address)
VALUES ('CUST001', 'John', 'john@gmail.com', '9876543210', 'New York');
```

```
INSERT INTO customer (SSN, Name, Email, Phone, Address)
VALUES ('CUST002', 'Alice', 'alice@gmail.com', '9123456780', 'Miami');
```

2. SELECT

Retrieves data from one or more tables

```
SELECT * FROM customer;
```

3. UPDATE

Modifies existing data in a table

Query

```
UPDATE customer
```

```
SET Phone = '900001234'
```

```
WHERE SSN = 'CUST001';
```


4. DELETE

Deletes one or more rows from a table

Query:

DELETE FROM ~~customer~~ WHERE SSN = 'CUST001'

VELTECH	
P.E. No.	2-1
PERFORMANCE (5)	5
SKILL AND ANALYSIS	5
VOICE (5)	5
RECORD (5)	15
TOTAL (20)	418
ON WITH DATE	

418

RESULT:

Thus all the DDL and DML commands are in SQL are executed successfully.

2.2 DDL and DML Commands with Constraints

AIM:

To perform DDL and DML commands with constraints in SQL

CONSTRAINTS

NOT NULL

UNIQUE

PRIMARY KEY

FOREIGN KEY

CHECK

DEFAULT

NOT NULL

It ensures a column cannot store NULL values

Syntax: CREATE TABLE TABLE_NAME (COLUMN NAME DATATYPE
NOT NULL);

UNIQUE

It ensures all values in a column are unique

Syntax:

CREATE TABLE table name (column name datatype UNIQUE);

PRIMARY KEY:

It is the combination of NOT NULL and UNIQUE

FOREIGN KEY

It ensures value in one table match value in another table

Syntax: FOREIGN KEY (column name) REFERENCE

DEFAULT:

It provides a default value for a column when none is specified.

DESC customer

Field	Type	Null	Key	Default
SSN	varchar(20)	NO	PR1	NULL
Name	varchar(100)	NO		NULL
Email	varchar(100)	YES	UNI	NULL
Phone	varchar(20)	YES		NULL
Address	varchar(200)	YES		UNKNOWN
DOB	date	NO		NULL

DDL commands with constraints

CREATE TABLE customer

SSN VARCHAR(20) PRIMARY KEY,

Name VARCHAR(100) NOT NULL,

E-mail VARCHAR(100) UNIQUE,

Phone VARCHAR(15) CHECK (LENGTH(phone)=10),

Address VARCHAR(100) DEFAULT 'unknown'

);

CREATE TABLE Hotel

HotelID INT PRIMARY KEY,

Name VARCHAR(20) NOT NULL,

Location VARCHAR(100) NOT NULL,

);

CREATE TABLE Rooms

RoomID INT PRIMARY KEY,

Name VARCHAR(50) NOT NULL,

HotelID INT,

);

ALTER TABLE:

~~ALTER TABLE~~ customer ADD DOB DATE NOT NULL;

ALTER TABLE customer MODIFY Phone VARCHAR(20);

TRUNCATE TABLE

TRUNCATE TABLE HOTEL;

RENAME TABLE

RENAME TABLE Customer to customers

After inserting data

SELECT * from customers

SSN	Name	Email	Phone	Address	DOB
CUST001	John	john@gmail.com	8987654321	Unknown	12-05-1990
CUST002	Alice	alice@gmail.com	9866543210	Unknown	06-10-1992
CUST003	Michael	mike@gmail.com	9765413264	Unknown	07-09-1995

After updating data

SELECT * from customers

SSN	Name	Email	Phone	Address	DOB
CUST001	John	john@gmail.com	8987654321	Unknown	12-05-1990
CUST002	Alice	alice@gmail.com	9866543210	Unknown	06-10-1992
CUST003	Michael	mike@gmail.com	9765413264	Unknown	07-09-1995

After deleting data

SELECT * from customers

SSN	Name	Email	Phone	Address	DOB
CUST001	John	john@gmail.com	8987654321	Unknown	12-05-1990
CUST002	Alice	alice@gmail.com	9866543210	Unknown	06-10-1992

SSN	Name
CUST001	John
CUST002	Alice

DML command with constraints

1. INSERT Data

```
INSERT INTO customers (SSN, Name, E-mail, Phone, Address, DOB)
VALUES ('CUST001', 'John', 'john@gmail.com', '9876543210', 'New York',
'12-05-1990');
```

```
INSERT INTO customers (SSN, Name, Email, Phone, Address, DOB)
VALUES ('CUST002', 'Alice', 'alice@gmail.com', '9866543210', 'Miami',
'06-10-1992');
```

```
INSERT INTO customers (SSN, Name, Email, Phone, Address, DOB)
VALUES ('CUST003', 'Michael', 'mike@gmail.com', '9765413264', 'London',
'07-09-1995');
```

2. UPDATE Data

```
UPDATE customers
SET E-mail = john@gmail.com
WHERE SSN = CUST001;
```

3. DELETE Data

```
DELETE FROM customers
WHERE SSN = 'CUST003';
```

4. SELECT DATA

```
SELECT SSN, Name
FROM customers;
```

VELTECH	
EX No.	22
PERFORMANCE (5)	5
RESULT AND ANALYSIS	5
VIVA VOCE (5)	5
RECORD (5)	15
TOTAL (70)	0
WITH DATE	11/8

RESULT
11/8

Thus all the DDL and DML commands with constraints are performed and executed successfully.